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RESEARCH NOTE

Teaching a Physical Activity to a Young Child With Autism Using the Three-Tier Video Modeling Model: A Pilot Study

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The Three-Tier Video Modeling intervention model was derived from the theory of Response to Intervention. This pilot study aimed to demonstrate the application of the Three-Tier Video Modeling model with a top-down approach in teaching physical activity to a 4-year-old boy with autism spectrum disorder and his typically developing peers. The study utilized a single case-changing conditions design and measured children's performance in the target physical activity. The results showed that the boy with autism spectrum disorder acquired the target skill and consequently generalized it to the natural environment, and all peers mastered the skill after intervention. Teachers and parents also reported a more inclusive learning environment. The potential implications of this novel approach were discussed.

Keywords: autism spectrum disorder, inclusive physical education, skill acquisition, generalization

Physical activity is essential for every child, including children with autism spectrum disorder (ASD). Previous researchers indicated that physical activities were associated with better physical health, social skills, motor skills, cognition, and behaviors for individuals with ASD (Tiner et al., 2021). However, children with ASD may have a broad range of social-communication deficits, repetitive behaviors, and/or challenging behaviors (Lord et al., 2018), making it challenging to teach physical activities to children with ASD. A study indicated several common barriers to inclusive physical education including teachers' negative attitudes, lack of knowledge of teaching students with disabilities, and unwelcoming peers (Haegele et al., 2021).

Video modeling (VM) is an evidence-based practice for teaching individuals with ASD (Steinbrenner et al., 2020). Research has shown that VM is effective in acquiring skills, increasing social interaction, and improving behavior problems for individuals with ASD and intellectual disability (Park et al., 2019; Steinbrenner et al., 2020). One of the theoretical frameworks underlying the use of VM is rooted in the social learning theory introduced by Bandura in 1977 (Case & Yun, 2018; Judge & Morgan, 2020). This theory explains how individuals learn by observing others and modeling transmitting behaviors. Four core principles of observational learning process are: attention, retention, reproduction, and motivation. The process of observational learning enables individuals to engage, imitate, and establish new behavior patterns.

VM aligns with the principles of the social learning theory and proves effective for teaching individuals with ASD due to its ability to address specific ASD characteristics. The researcher explains how VM could improve attention and motivation processes through addressing ASD characteristics including over-selective attention, a preference for visual stimuli, avoidance of eye contact, and a restricted field of focus (Corbett et al., 2005). Additionally, VM aids in retention through task analysis and repetitive exposure, enabling individuals with ASD to decode and memorize behavior patterns (Corbett et al., 2005; Judge & Morgan, 2020).

Some research has found that VM is effective in teaching motor skills to individuals with ASD. For example, Cannella-Malone et al. (2013) used VM to teach and foster leisure skills (e.g., jumping ropes and ladder drills) for three adolescents with ASD or intellectual disabilities. In a more recent study, researchers compared the effectiveness of VM, VM with point-light model, and control groups in learning basketball shots for 45 adolescents with ASD. Findings suggest that both VM and VM with point-light model were more effective than the traditional teaching procedure (Baniasadi et al., 2022). Nevertheless, research applying VM in physical education is relatively scarce, especially for young children with ASD.

A Tiered Model to Implement VM

There are several types of VM, such as others modeling (e.g., peers or adults), video self-modeling (VSM), and point-of-view VM. VM using others as models or point-of-view is easy to generate and access as generic videos in which the learner neither recognizes the model nor the setting, and so on. On the other hand, VSM is the most individualized since it uses the target learner as the model performing a target skill or behavior in the identical setting (Steinbrenner et al., 2020). Some studies suggest VSM is particularly robust in teaching children with ASD because of its nature of individualization and the potential to facilitate self-efficacy (Buggey & Ogle, 2012; Hansen et al., 2022; Zarate et al., 2021). However, creating a VSM can take much more time compared with other types of VM and usually requires video editing techniques, making it challenging for teachers to apply or sustain (Buggey & Ogle, 2012; Van Laarhoven et al., 2009).

To address the sustainability of VM in the inclusive educational setting, Wang and Koyama (2014) developed the Three-Tier VM (3TVM) intervention model based on the logic of the Response to Intervention (RTI) model (Searle, 2010). The RTI model is a three-tiered system of support for students with special needs which helps teachers better identify and meet students' needs with more efficient resource allocation. Inspired by the RTI model,

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the 3TVM also consists of three tiers: primary, secondary, and tertiary tiers (Figure 1).

The 3TVM can be applied in two directions: top-down or bottom-up (Wang, 2016). In the bottom-up approach, the primary tier is initially introduced to all learners

using generic VM. Educators then identify those learners who do not respond to the primary tier and implement the secondary tier to provide additional support with the small group VM instruction and feedback. For individuals requiring more intensive support, educators offer the tertiary tier intervention, which typically involves individualized VSM and personalized strategies including preference assessment, reinforcement, prompting procedures, specific praise, and feedback.

Contrarily, the top-down approach begins with the identification of individuals with special needs. For example, a child has been diagnosed with ASD and an individualized education plan has been developed. In this approach, the tertiary tier intervention is implemented first, followed by the secondary and primary tiers. This sequence aims to help the learner acquire the skill and gradually generalize it in a natural setting alongside peers. In our study, we employed the top-down approach for a child identified with ASD, assisting him in learning a physical activity and generalizing the skill in a natural setting.

Although the RTI model has been implemented across the United States for a decade and its logic has been incorporated into varied intervention or training (e.g., Berkeley et al., 2020; Canfield et al., 2023), the use of 3TVM is relatively new and more research is needed to examine its effectiveness and utility.

Research Questions

The present study aimed to demonstrate how VM can be implemented with 3TVM in inclusive physical education and investigate its potential effects in promoting skill acquisition and generalization. The novel application was piloted with a 4-year-old preschool child with ASD and his peers learning a physical activity: walking on tin-can stilts. Specifically, the research questions were: (a) Will the child with ASD successfully master the target activity following the tertiary tier intervention? (b) Will the child with ASD be able to generalize the learned skill to a small group setting, and then to the entire class?

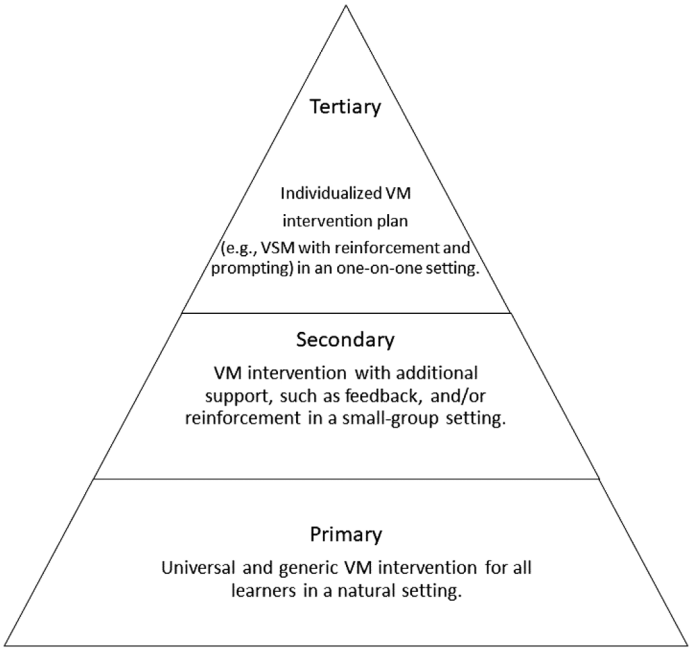


Figure 1 — Illustration of 3TVM. *Note.* VM = video modeling; VSM = video self-modeling. Adapted from "An Analysis and Review of the Literature and a Three-Tier Video Modeling Intervention Model," by H.-T. Wang and T. Koyama, 2014, *Research in Autism Spectrum Disorders, 8*(7), pp. 746–758. Copyright 2014 from Elsevier.

Methods

Participants

The Target Participant

A 4-year-old boy with ASD, named Chan, was identified with the following inclusion criteria: the child could (a) imitate basic gross motor and verbal skills, such as "touch head" or saying "touch head"; (b) speak in simple sentences; (c) demonstrate picture matching skill; (d) recognize oneself in a video; (e) watch 3-min video clips; and (f) follow simple instructions, such as "do this" or "raise your hand." Chan was assessed using the Peabody Developmental Motor Scales (second edition; Folio & Fewell, 2000), which indicated developmental delays in fine motor and gross motor areas, but the motor delay did not prevent his participation in physical activities. While Chan was able to imitate words verbally, he had a limited vocabulary and little interaction with peers. He often wandered around and did not engage in activities.

Peers

Three 4-year-old peers in Chan's preschool class, two girls and one boy, were invited to participate in the secondary tier of 3TVM intervention. These children were selected because they often initiated social interactions or played with Chan. During the primary tier of 3TVM intervention, 27 peers aged between 4 and 5 years in Chan's class participated in this research. Among these participants, one participant was diagnosed with motor developmental delay, while others were neurotypical children. We obtained informed consent from all participants' legal guardians before the research.

The Target Physical Activity

The target physical activity was walking on tin-can stilts, which is a popular physical activity in preschools in Taiwan. This activity was chosen because it was included in the target participant's individualized education plan and was recommended in the preschool curriculum guideline for physical balance and coordination. Task analysis was used to break down the activity into 11 steps. To complete the 11 steps, the child should take a pair of tin-can stilts out, pick up strings, stand on the tin cans, walk on the tin-can stilts with feet moving reciprocally, walk to the first cone, walk around the first cone, walk to the second cone, walk around the second cone, stop at a squared area, lower feet down to the ground, and put the stilts back. The total distance was 6 m. The average duration of completing the activity was 10–12 min. Each step was operationally and clearly defined. For example, during Step 3, the child stands with two feet on each tin-can stilt.

Modeling Video

This study developed multiple exemplar modeling videos consisting of one self-modeling and three peer-modeling segments.

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To create the self-modeling segment, the researcher selected clips from the baseline recording that depicted Chan successfully performing certain steps, and the prompts or assistance was subsequently edited out. The resulting segments were combined to produce a video in which Chan successfully and independently demonstrated all the steps of the target activity. To generate three segments of peer-modeling, three peers of Chan were recorded performing 11 steps of walking on tin-can stilts. By combining all segments, a four-exemplar video was generated and concluded with praise and tangible reinforcement for each model. The total duration of the video was about 8 min.

Dependent Variable

As described previously, the target activity, walking on stilts, was divided into 11 steps. The outcome variable measured how many steps participants could perform independently and correctly. Visual analysis was used to analyze the behavioral data in each tier, including descriptions of changes in data level, trend, variability, and directions of each participant.

Research Design

This pilot study adopted a single case changing conditions design (e.g., McIntyre & Phaneuf, 2008) consisting of the following phases: baseline, tertiary tier, secondary tier, and primary tier. Before baseline, a preference assessment was done indirectly by interviewing Chan's parents and by the researcher's observation. Based on the assessment, some preferred edibles and toys were presented on a goodie tray for reinforcement.

Three-Tier Procedure

During baseline, the researcher asked the participant to walk on stilts by saying "Chan, let's walk on the stilts" without providing further prompts or instructions. If the participant did not respond within 3 s, the researcher repeated the instruction and started a 40-s timer. After 40 s elapsed, if the participant did not respond, the session ended. If the participant engaged in the activity, the researcher recorded the number of steps that the participant independently performed. The baseline phase ended when the data points demonstrated a stable trend and level.

The intervention began with the tertiary tier. In this tier, the researcher conducted one-on-one sessions. Sessions ended within 20 min. First, Chan watched the modeling video in a quiet library without receiving additional prompts or instructions. While the video was on, Chan could walk around but was praised if he watched the video. Then, the researcher took Chan to an indoor gym and asked Chan to walk on stilts using the same procedure implemented in the baseline. If Chan independently and correctly performed all 11 steps, the research provided social praise and preferred items from the goodie tray. After Chan performed 11 steps for three consecutive sessions, the researcher gradually faded the reinforcement by offering fewer reinforcement items until all reinforcement items were removed. The tertiary tier ended when Chan continued performing 11 steps without reinforcement.

During the secondary tier, the researcher conducted small group sessions in which Chan watched the video with three peers in a quiet space. Then, the researcher asked Chan and his peers to walk on stilts together in the indoor gym using the same procedure implemented in the baseline. Children who completed 11 steps received praise and items from the goodie tray. The reinforcement faded when all children in this small group performed 11 steps for at least three consecutive sessions. The secondary tier ended when all children performed 11 steps without reinforcement.

Before implementing the primary tier intervention, the researcher collected baseline data for all children in Chan's classroom using the procedure described in the baseline phase. After data points of the class average demonstrated stable trends and levels, the researcher implemented the primary tier intervention in Chan's classroom. All 27 children in Chan's class participated in the primary tier. First, the class watched the video together in the classroom. During the video, the researcher did not provide additional instructions. Then, the researcher asked the class to walk on stilts together in the corridor where they usually play stilts during recess. Children who completed 11 steps received praise and reinforcement. The reinforcement faded when the whole class performed 11 steps for three consecutive sessions. The primary tier ended when the class performed 11 steps without reinforcement.

The maintenance effect was evaluated at 1-week and 1-month follow-ups. Two follow-up sessions were conducted for all children in Chan's classroom during recess in the corridor. During follow-up sessions no intervention or reinforcement was delivered. All children performed in the natural setting.

Reliability

A trial-by-trial interobserver agreement and intervention fidelity were conducted during 50% of sessions in each tier by two independent observers. The secondary observer was blind to research phases and design but was trained with sample session videos before recording the data. The average interobserver agreement reached 94.6% (ranged 81%–100%) and fidelity reached 98.3% (ranged 75%–100%).

Social Validity

Social validity was evaluated using inclusion indicators proposed by Heron and Skinners (1981). The four indicators include social interaction, language communication, emotional expression, and peer acceptance. Teachers and parents reported positive changes in all four indicators, indicating an improvement in inclusiveness. Teachers observed that peers initiated more social interactions, such as inviting Chan to participate in various activities, and showed more positive comments and emotional expressions, such as cheering for Chan during playtime. Chan also engaged in more spontaneous conversation.

Results

Tertiary Tier

Figure 2 shows the changes in the number of independent steps. During baseline, Chan demonstrated zero to seven steps, with an average of four steps. After the intervention began, there was an immediate increasing trend in the number of independent steps from five to nine. Chan achieved 11 correct steps after four sessions. Although he dropped to nine steps after sick leave, he immediately increased independent steps to 11 steps the next day and maintained the performance. The researcher faded the goodie tray after Chan achieved 11 steps for 3 consecutive days and Chan continuously performed all 11 steps independently. See Figure 2 for the independent steps of Chan and peers.

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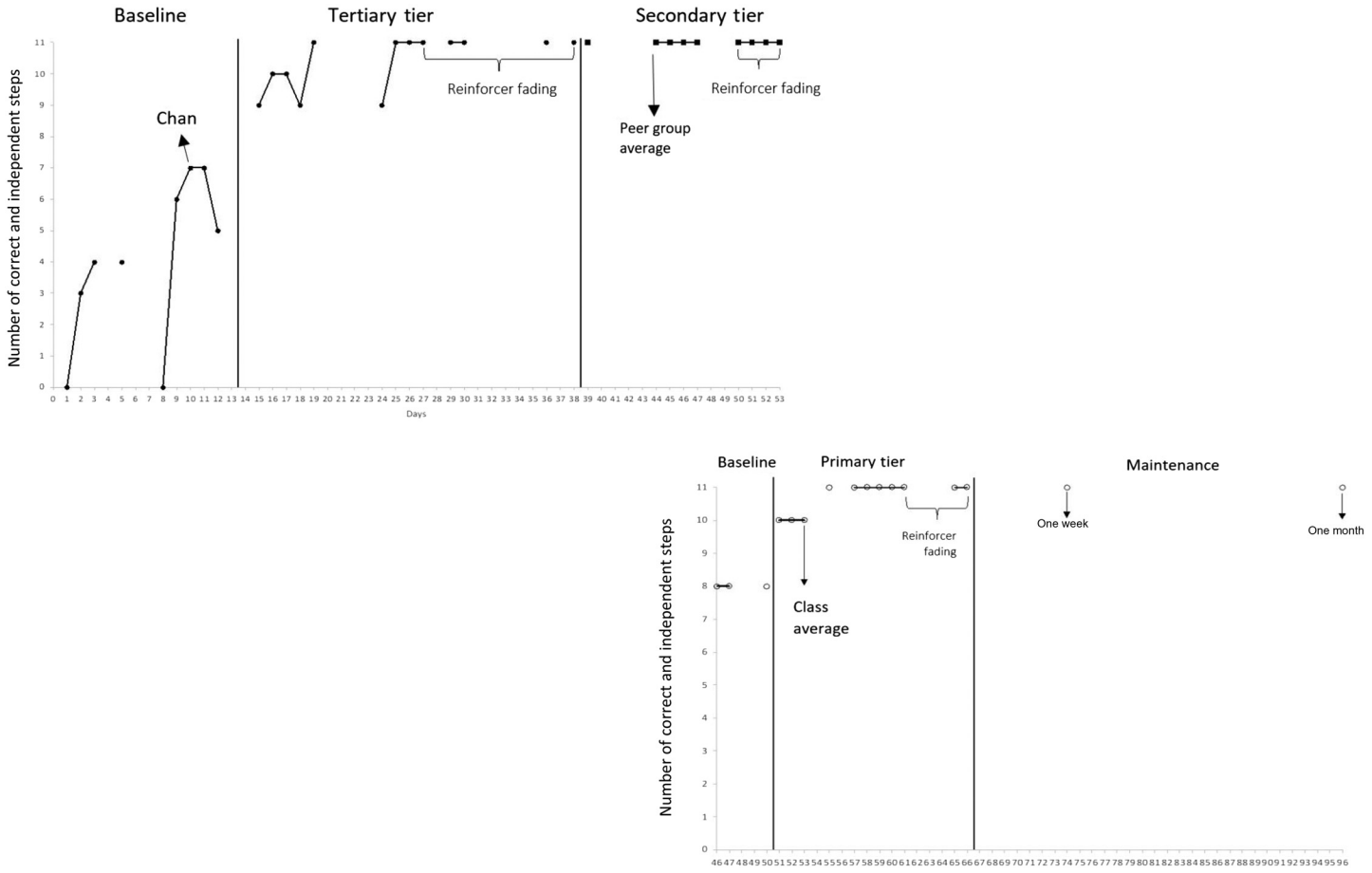


Figure 2 — Number of correct and independent steps.

Secondary Tier

During the secondary tier, Chan maintained the performance and generalized it from an individual training setting to a small group setting. Chan and three selected peers successfully performed all 11 steps in the secondary tier for five sessions. After the researcher withdrew the reinforcement, Chan and his peers continuously maintained 11 steps in the small group setting.

Primary Tier

Before conducting the primary tier intervention, the researcher collected the baseline of Chan and his classmates. The average number of steps in the baseline phase of the primary tier was eight for all participants, including Chan and his classmates ($n = 28$), ranging from six to 11 steps. During the intervention, there was an increasing trend for the average number of steps from eight to 11 steps. The average number of steps increased to 10 steps immediately after the intervention started, ranging from 10 to 11 steps. All children in Chan's class performed the 11 steps independently within four intervention sessions and continued to perform 11 steps for three sessions when the reinforcement faded.

Maintenance

During the maintenance phase, the entire class achieved 11 steps independently 1 week and 1 month after the intervention finished, showing that all children including Chan had mastered and maintained the performance in the natural learning environment.

Discussion

This pilot study demonstrated the application of VM with the 3TVM top-down approach on a 4-year-old boy with ASD and his peers in an inclusive physical education setting. Findings suggest that this novel approach was effective in teaching, generalizing, and maintaining skills of physical activity for the participant. The preliminary results support that VM could be used in inclusive physical education to improve skill acquisition for children with ASD (Cannella-Malone et al., 2013; Park et al., 2019). The study also highlights the utility of 3TVM to achieve treatment outcomes and increase sustainability by reusing the video systematically and generalizing the effectiveness to the whole class, which may minimize teachers' burden of editing videos and increase their use of VM in the inclusive classroom setting.

Implications

Implications of this study can help address some common barriers to inclusive physical education (Haegele et al., 2021). First, previous research shows the effectiveness of VM in teaching motor skills to individuals with ASD, yet most research focuses on adolescents or adults with ASD (e.g., Baniasadi et al., 2022;

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Bassetter et al., 2021). This study expands the literature on using VM to teach motor skills to young children with ASD, suggesting VM is another effective strategy for physical education teachers to consider when teaching young learners with ASD.

Second, VM has advantages in increasing learners' engagement (Lima et al., 2020; Steinbrenner et al., 2020) and aligns the principles of the observational learning process of the social learning theory (Baniasadi et al., 2022; Case & Yun, 2018). In the present study, the use of VSM and reinforcement procedures further addressed the attention and motivation process for the young learner with ASD. The video not only showed the target child successfully performing the skill but also showed that he received positive reinforcement after completing the task. As suggested in previous studies, these individualized features may contribute to the learning effects for individuals with ASD (Bassetter et al., 2021; Hansen et al., 2022; Zarate et al., 2021).

Third, the video used in the present study included four segments with one segment of VSM and three segments of peer-modeling videos. All segments depicted the same steps of the activity. Such repetitive demonstration of the steps is essential for facilitating the retention process for children with ASD (Corbett et al., 2005; Judge & Morgan, 2020; Wang & Koyama, 2014). Providing learners with opportunities to perform the skill right after VM implementation is recommended as it is part of the reproduction process of observational learning (Judge & Morgan, 2020).

Fourth, the sustainability of VM in educational setting can be enhanced through the 3TVM top-down approach (Wang & Koyama, 2014). The VSM created in the tertiary tier for the learner with ASD can be served as an individualized peer-modeling video useful for teaching other typically developing peers who recognize the target child. Additionally, it can also potentially serve as a generic modeling video for those who do not recognize the target child as the model and are unfamiliar with the setting in the video.

Since it usually requires more time and advanced video editing techniques to create VSM (Buggey & Ogle, 2012; Van Laarhoven et al., 2009), practitioners may consider developing the tertiary intervention by incorporating personalized behavior support strategies (e.g., reinforcement procedure) with other forms of VM that are easier to generate and individualized, such as point-of-view VM.

Finally, teachers and parents in this study observed that the top-down 3TVM approach may facilitate social acceptance and positive interactions among the target child and their peers. Our report on social validity using inclusion indicators (Heron & Skinners, 1981) suggested that the use of VSM with the 3TVM model may potentially support overall inclusiveness in the classroom and help increase the awareness of inclusion. Future studies can measure inclusiveness directly with more comprehensive inclusion assessment tools (e.g., Knickenberg et al., 2020).

Limitations

Since it was a pilot study, some issues should be noted in this study. First, this study may not indicate a strong functional relationship between 3TVM and the outcomes because of the nature of the research design. However, the research design reserves the functions of prediction, verification, and replication (Alberto & Troutman, 2013, p. 209). Future research on a larger scale is needed. Second, the top-down approach may be more appropriate for some educational contexts or systems. The context should be considered to determine whether to use the top-down or the bottom-up approach. Practitioners should also consider their available resources and support to determine which direction of approach is more feasible.

Conclusions

This pilot study supported the use of VM in teaching motor skills to young children with ASD in the inclusive physical education. The study demonstrated how 3TVM model with the top-down approach could help children with ASD to gradually generalize the motor skill to a natural setting and how VM can be implemented with sustainability.

Compliance With Ethical Standards

On behalf of both authors, the corresponding author states that there is no conflict of interest. This study involved human participants and was performed in line with the 1964 Declaration of Helsinki and later comparable ethical standards of the authors' research institute, the National Taiwan Normal University. Informed consent was obtained from the participant's parents or legal guardians before the research began.

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